# Technical Report: Empirical Evaluation of Initial-Condition Scaling and Plateau-Based Neighborhood Selection in Local PCA for Müller–Israel–Stewart Hydrodynamics

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**Abstract**

This report provides a formal mathematical specification and numerical execution report for estimating the local intrinsic dimension of non-equilibrium solutions in Conformal Müller–Israel–Stewart (MIS) hydrodynamics under boost-invariant Bjorken expansion. We implement and evaluate four data normalization procedures: (1) Variant 1, in which scale factors for temperature and pressure anisotropy are frozen from the initial condition standard deviations at $t_0$; (2) Variant 2, in which transverse anisotropy scaling remains frozen at $t_0$ while longitudinal temperature scaling accounts for Bjorken expansion cooling; (3) Variant 3, formulated using the logarithm of temperature with frozen initial scales; and (4) a reference benchmark employing standard slice-by-slice $Z$-score standardization. In each case, local neighborhoods are identified via $k$-nearest neighbors in the scaled metric, empirical covariance matrices are computed, and spectral properties are evaluated using the continuous Participation Ratio ($d_{\mathrm{PR}}$). In addition, we execute a finite-difference plateau-detection algorithm to locate the inflection point $k^*$ where statistical undersampling stabilizes prior to the onset of curvature contamination. All mathematical equations, numerical configurations, phase-space distributions, eigenvalue trajectories, and dimension curves are presented as executed.

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# 1 Hydrodynamic Setup and Initial Phase-Space Configuration

We consider a system of boost-invariant, 1D longitudinally expanding quark-gluon plasma described by the conformal Müller–Israel–Stewart (MIS) hydrodynamic equations. The state of the fluid at proper time $\tau$ is specified by the two-dimensional physical state vector:

$$\mathbf{x}(\tau) = \begin{pmatrix} T(\tau) \\ \mathcal{A}(\tau) \end{pmatrix}, \tag{1}$$

where $T(\tau)$ denotes the local temperature (expressed in $\mathrm{fm}^{-1}$, with $1\,\mathrm{fm}^{-1} \approx 197.3\,\mathrm{MeV}$), and $\mathcal{A}(\tau) \equiv (\mathcal{P}_T - \mathcal{P}_L)/\mathcal{P}$ represents the dimensionless pressure anisotropy.

## 1.1 Equations of Motion

From the conservation of energy-momentum $\partial_\mu T^{\mu\nu} = 0$, the temperature evolves according to the exact Bjorken cooling relation:

$$\tau \frac{dT}{d\tau} = T\left(-\frac{1}{3} + \frac{\mathcal{A}}{18}\right). \tag{2}$$

For conformal fluids, the pressure anisotropy relaxes toward the universal hydrodynamic attractor $\mathcal{A}_{\mathrm{attr}}(w)$ via:

$$\tau \frac{d\mathcal{A}}{d\tau} = \frac{1}{C_{\tau_\pi}}\left[8C_\eta - w\mathcal{A} - \frac{2}{9}C_{\tau_\pi}\mathcal{A}^2\right], \tag{3}$$

where $C_\eta \equiv \eta/s$, $C_{\tau_\pi} \equiv \tau_\pi T$, and $w \equiv \tau T$ is the dimensionless scaling variable. Linearization around the attractor yields the transverse relaxation rate:

$$\tau \frac{d\mathcal{A}}{d\tau} \approx -\gamma(w)\big(\mathcal{A} - \mathcal{A}_{\mathrm{attr}}(w)\big), \qquad \gamma(w) \equiv \frac{w}{C_{\tau_\pi}} = \frac{\tau}{\tau_\pi(T)} \gg 1. \tag{4}$$

## 1.2 Numerical Dataset and Initial Reference Scales

The dataset analyzed consists of $N = 10\,000$ independent trajectories integrated from initial proper time $\tau_0 = 0.20\,\mathrm{fm}/c$ up to $\tau_{\max} = 3.00\,\mathrm{fm}/c$ sampled across 281 proper-time slices. The initial conditions at $\tau_0$ are distributed across:

$$T_i(\tau_0) \in [1.0136,\, 6.0811]\,\mathrm{fm}^{-1}, \qquad \mathcal{A}_i(\tau_0) \in [0.0003,\, 5.9996]. \tag{5}$$

At the initial reference slice $\tau_0 = 0.20\,\mathrm{fm}/c$, the sample statistics evaluated across the ensemble are:

$$\langle T_0 \rangle = \frac{1}{N}\sum_{i=1}^{N} T_i(\tau_0) = 3.5528\,\mathrm{fm}^{-1}, \tag{6}$$

$$x_T(t_0) \equiv \sigma_T(\tau_0) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\big(T_i(\tau_0) - \langle T_0 \rangle\big)^2} = 1.4513\,\mathrm{fm}^{-1}, \tag{7}$$

$$x_A(t_0) \equiv \sigma_{\mathcal{A}}(\tau_0) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\big(\mathcal{A}_i(\tau_0) - \langle \mathcal{A}_0 \rangle\big)^2} = 1.7219, \tag{8}$$

$$x_{\ln T}(t_0) \equiv \sigma_{\ln T}(\tau_0) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(\ln\frac{T_i(\tau_0)}{\langle T_0 \rangle}\right)^2} = 0.4729. \tag{9}$$

# 2 Coordinate Normalization Schemes

To evaluate distances between states with different physical units ($T$ in $\mathrm{fm}^{-1}$, $\mathcal{A}$ dimensionless), we define a diagonal scaling transformation $\mathbf{S}^{-1}(t)$.

### 2.1 Variant 1: Frozen Scale Normalization

In Variant 1, the scaling matrix is fixed permanently to the empirical standard deviations measured at $t_0$:

$$\mathbf{S}_0^{-1} = \begin{pmatrix} 1/x_T(t_0) & 0 \\ 0 & 1/x_A(t_0) \end{pmatrix}. \tag{10}$$

The scaled coordinate vector for trajectory $i$ at time $t \geq t_0$ is defined by:

$$\tilde{\mathbf{x}}_i^{(1)}(t) = \begin{pmatrix} \tilde{T}_i(t) \\ \tilde{\mathcal{A}}_i(t) \end{pmatrix} = \begin{pmatrix} T_i(t)/x_T(t_0) \\ \mathcal{A}_i(t)/x_A(t_0) \end{pmatrix}. \tag{11}$$

The squared distance between two states $i$ and $j$ in this metric is:

$$d_{ij}^2(t) = \left( \frac{T_i(t) - T_j(t)}{x_T(t_0)} \right)^2 + \left( \frac{\mathcal{A}_i(t) - \mathcal{A}_j(t)}{x_A(t_0)} \right)^2. \tag{12}$$

At $t = t_0$, the global covariance matrix is by construction the identity matrix:

$$\tilde{\mathbf{\Sigma}}_{\text{glob}}(t_0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{13}$$

For $t > t_0$, the transverse thickness contracts as $\delta\mathcal{A}_\perp(t) \sim x_A(t_0) e^{-\int_{t_0}^t \gamma dt'}$. Because the divisor $x_A(t_0)$ is frozen, the transverse variance decreases toward zero:

$$\lambda_2(t) \approx \left( \frac{\delta\mathcal{A}_\perp(t)}{x_A(t_0)} \right)^2 \xrightarrow{t \gg t_0} 0. \tag{14}$$

However, energy conservation in Bjorken expansion implies $T_i(t) \approx T_i(t_0)(t_0/t)^{1/3}$. Therefore, the temperature spread shrinks kinematically as:

$$T_i(t) - T_j(t) \approx \big(T_i(t_0) - T_j(t_0)\big) \left( \frac{t_0}{t} \right)^{1/3}, \tag{15}$$

which causes the global variance of the first coordinate in Variant 1 to contract as:

$$\mathrm{Var}\big(\tilde{T}(t)\big) = \mathrm{Var}\left( \frac{T(t)}{x_T(t_0)} \right) \approx \left( \frac{t_0}{t} \right)^{2/3} \xrightarrow{t \to \infty} 0. \tag{16}$$

### 2.2 Variant 2: Cooling-Compensated Scale Normalization

In Variant 2, the temperature divisor is scaled dynamically by the mean ensemble cooling factor, while the transverse divisor $x_A(t_0)$ remains frozen:

$$x_T(t) \equiv x_T(t_0) \left( \frac{\langle T(t) \rangle}{\langle T(t_0) \rangle} \right), \qquad x_A = x_A(t_0). \tag{17}$$

The scaled coordinates are given by:

$$\tilde{\mathbf{x}}_i^{(2)}(t) = \begin{pmatrix} T_i(t)/x_T(t) \\ \mathcal{A}_i(t)/x_A(t_0) \end{pmatrix}. \tag{18}$$

Under this formulation, the ensemble variance along the longitudinal temperature direction is maintained at $\mathrm{Var}(\tilde{T}(t)) \approx 1$ for all times $t \geq t_0$.

### 2.3 Variant 3: Logarithmic Temperature Normalization

In Variant 3, the temperature coordinate is expressed in logarithmic form with frozen scales:

$$\tilde{\mathbf{x}}_i^{(3)}(t) = \begin{pmatrix} \tilde{y}_{T,i}(t) \\ \tilde{\mathcal{A}}_i(t) \end{pmatrix} = \begin{pmatrix} \ln\big(T_i(t)/\langle T_0\rangle\big)/x_{\ln T}(t_0) \\ \mathcal{A}_i(t)/x_A(t_0) \end{pmatrix}. \tag{19}$$

Because $\ln T_i(t) \approx \ln T_i(t_0) - \frac{1}{3}\ln(t/t_0)$, the cooling term $-\frac{1}{3}\ln(t/t_0)$ shifts all trajectory points by an identical scalar translation. Consequently, the variance $\mathrm{Var}(\tilde{y}_T(t)) \approx 1$ is preserved without requiring explicit time-dependent rescaling.

### 2.4 Reference Benchmark: Slice-by-Slice $Z$-Score Standardization

As a reference comparison, we evaluate standard per-slice $Z$-score standardization, where the sample mean $\mu(t)$ and sample standard deviation $\sigma(t)$ are recomputed at each time slice independently:

$$\tilde{\mathbf{x}}_i^{(\mathrm{Z})}(t) = \begin{pmatrix} \big(T_i(t) - \mu_T(t)\big)/\sigma_T(t) \\ \big(\mathcal{A}_i(t) - \mu_{\mathcal{A}}(t)\big)/\sigma_{\mathcal{A}}(t) \end{pmatrix}. \tag{20}$$

Under this transformation, both coordinate axes are forced to unit variance ($\sigma_{\tilde{T}} = \sigma_{\tilde{\mathcal{A}}} = 1$) at every time slice $t$.

### 2.5 Phase-Space Scatter Evolution

Figure 1 displays the empirical distributions of the trajectory cloud in scaled phase space $(\tilde{x}_1, \tilde{x}_2)$ across four proper-time slices: $\tau = 0.20\,\mathrm{fm}/c$, $\tau = 0.45\,\mathrm{fm}/c$, $\tau = 0.85\,\mathrm{fm}/c$, and $\tau = 2.00\,\mathrm{fm}/c$. Each row corresponds to one of the four normalization schemes, and the markers are color-coded by the initial anisotropy $\mathcal{A}_i(\tau_0)$.

## 3 Local PCA and Spectral Eigenvalue Dynamics

For each trajectory point $\tilde{\mathbf{x}}_i(t)$ in scaled space, we identify its $k$ nearest neighbors $\mathcal{N}_k(\tilde{\mathbf{x}}_i)$ using a $k$-d tree with standard Euclidean metric:

$$\mathcal{N}_k(\tilde{\mathbf{x}}_i) = \{\tilde{\mathbf{x}}_j \mid \|\tilde{\mathbf{x}}_j - \tilde{\mathbf{x}}_i\| \le r_k(\tilde{\mathbf{x}}_i)\}, \qquad |\mathcal{N}_k(\tilde{\mathbf{x}}_i)| = k. \tag{21}$$

The local sample mean and empirical covariance matrix are:

$$\bar{\mathbf{x}}_i = \frac{1}{k}\sum_{j\in\mathcal{N}_k(i)} \tilde{\mathbf{x}}_j, \qquad \mathbf{C}_i = \frac{1}{k-1}\sum_{j\in\mathcal{N}_k(i)} (\tilde{\mathbf{x}}_j - \bar{\mathbf{x}}_i)(\tilde{\mathbf{x}}_j - \bar{\mathbf{x}}_i)^T \in \mathbb{R}^{2\times 2}. \tag{22}$$

The eigenvalues $\lambda_1^{(i)} \ge \lambda_2^{(i)} \ge 0$ of $\mathbf{C}_i$ are obtained by spectral decomposition:

$$\mathbf{C}_i \mathbf{v}_m^{(i)} = \lambda_m^{(i)} \mathbf{v}_m^{(i)}, \qquad m \in \{1, 2\}. \tag{23}$$

Here, $\lambda_1$ corresponds to the local sample variance along the principal direction (tangent to the attractor curve at late times), and $\lambda_2$ corresponds to the transverse variance normal to the curve.

### 3.1 Continuous Participation Ratio Dimension

The local Participation Ratio dimension $d_{\mathrm{PR}}(\tilde{\mathbf{x}}_i)$ is evaluated without spectral cutoffs:

$$d_{\mathrm{PR}}(\tilde{\mathbf{x}}_i) = \frac{(\mathrm{Tr}\,\mathbf{C}_i)^2}{\mathrm{Tr}\left(\mathbf{C}_i^2\right)} = \frac{\left(\lambda_1^{(i)} + \lambda_2^{(i)}\right)^2}{\left(\lambda_1^{(i)}\right)^2 + \left(\lambda_2^{(i)}\right)^2} = 1 + \frac{2\lambda_1^{(i)}\lambda_2^{(i)}}{\left(\lambda_1^{(i)}\right)^2 + \left(\lambda_2^{(i)}\right)^2}. \tag{24}$$

The ensemble mean dimension and spectral ratio are:

$$\langle d_{\mathrm{PR}}\rangle = \frac{1}{N}\sum_{i=1}^{N} d_{\mathrm{PR}}(\tilde{\mathbf{x}}_i), \qquad \langle \lambda_2/\lambda_1\rangle = \frac{1}{N}\sum_{i=1}^{N} \frac{\lambda_2^{(i)}}{\lambda_1^{(i)}}. \tag{25}$$

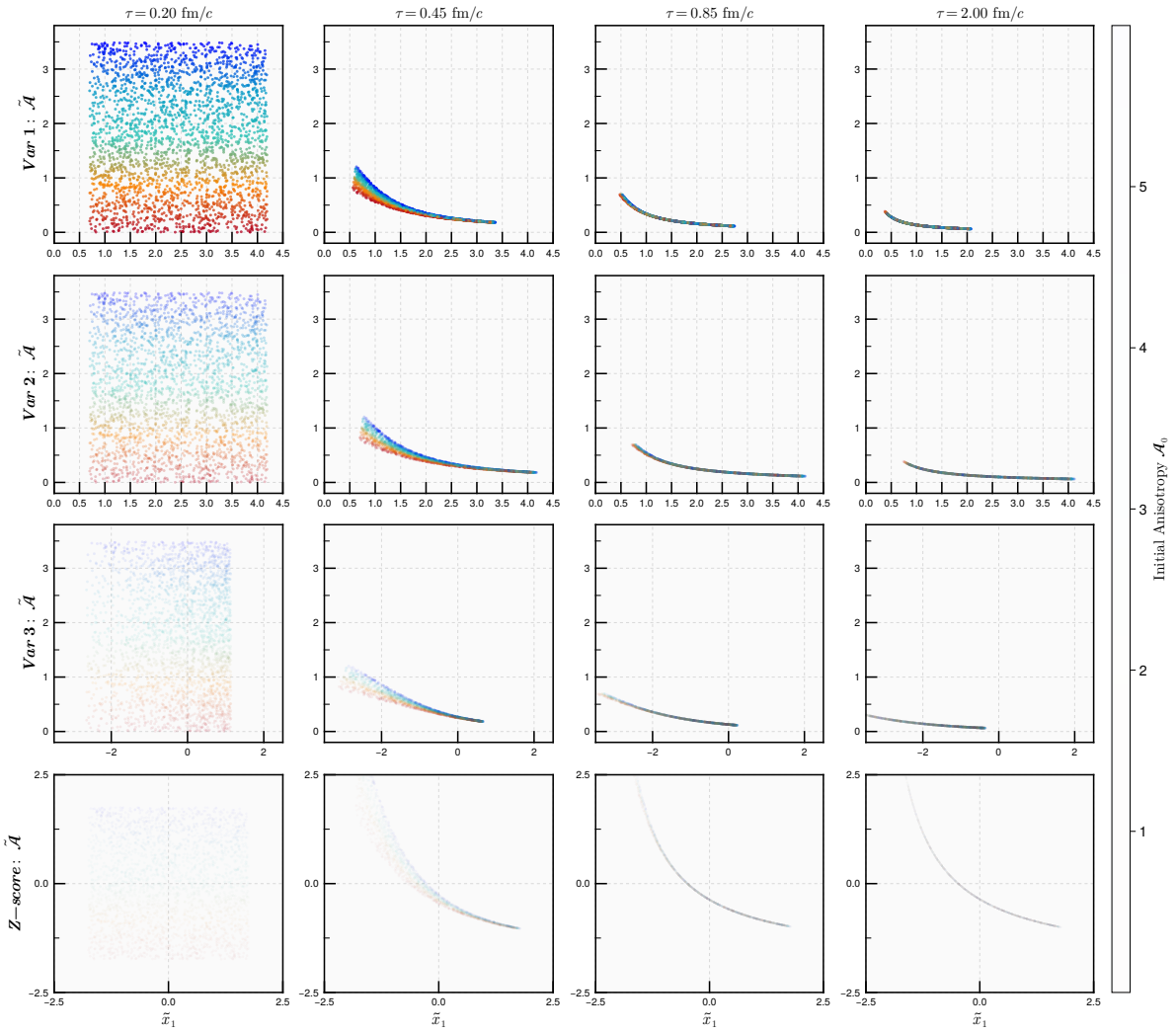



Figure 1: **Evolution of the trajectory ensemble in scaled phase space.** Slices at proper times $\tau \in \{0.20, 0.45, 0.85, 2.00\}\,\mathrm{fm}/c$ are displayed for Variant 1 (Row 1), Variant 2 (Row 2), Variant 3 (Row 3), and Per-Slice $Z$-Score (Row 4). Points are colored according to initial anisotropy $\mathcal{A}_i(\tau_0) \in [0, 6]$.

### 3.2 Eigenvalue Trajectories Across Proper Time

Figure 2 presents the trajectories of the ensemble-averaged eigenvalues $\langle\lambda_1\rangle$, $\langle\lambda_2\rangle$, and the spectral ratio $\langle\lambda_2/\lambda_1\rangle$ as a function of proper time $\tau \in [0.20, 3.00]\,\mathrm{fm}/c$ evaluated at fixed $k = 16$.

- **Panel (a):** In Variant 1, $\langle\lambda_1\rangle$ decreases from $1.88 \times 10^{-3}$ at $\tau = 0.20\,\mathrm{fm}/c$ to $5.51 \times 10^{-7}$ at $\tau = 3.00\,\mathrm{fm}/c$, matching the theoretical Bjorken cooling curve $\propto (\tau_0/\tau)^{2/3}$. In Variant 2 and Variant 3, $\langle\lambda_1\rangle$ stabilizes at asymptotic values of $2.65 \times 10^{-6}$ and $4.21 \times 10^{-6}$. In the per-slice $Z$-score benchmark, $\langle\lambda_1\rangle$ remains at $\approx 1.89 \times 10^{-5}$.
- **Panel (b):** In all frozen-scale methods (Variants 1, 2, and 3), the transverse eigenvalue $\langle\lambda_2\rangle$ decays from $1.03 \times 10^{-3}$ at $\tau = 0.20\,\mathrm{fm}/c$ to $7.13 \times 10^{-12}$ at $\tau = 3.00\,\mathrm{fm}/c$. In the per-slice $Z$-score benchmark, $\langle\lambda_2\rangle$ reaches $1.22 \times 10^{-10}$.
- **Panel (c):** The spectral ratio $\langle\lambda_2/\lambda_1\rangle$ decays from $\approx 0.57$ at $\tau_0$ to $7.99 \times 10^{-6}$ (Var 1), $2.85 \times 10^{-6}$ (Var 2), $5.51 \times 10^{-7}$ (Var 3), and $1.31 \times 10^{-6}$ ($Z$-score) at $\tau = 3.00\,\mathrm{fm}/c$.

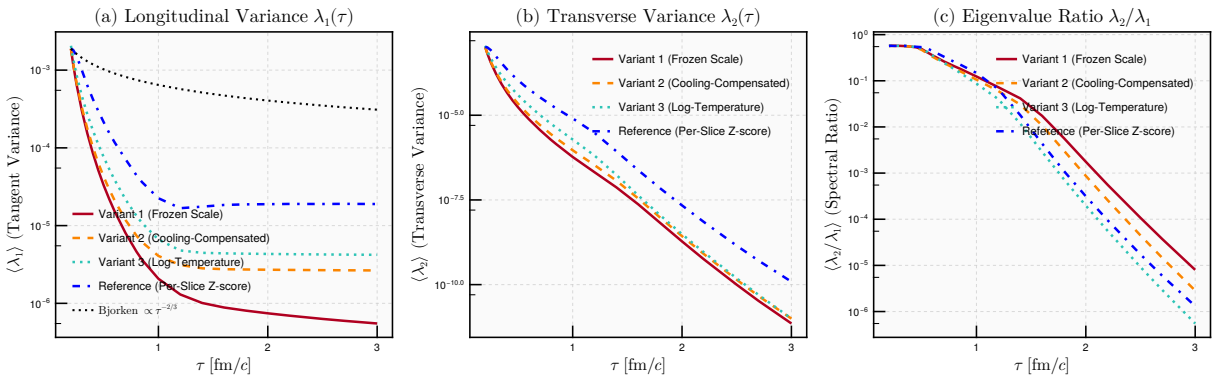



Figure 2: **Local PCA eigenvalue dynamics across proper time $\tau$ ($k = 16$).** (a) Mean principal eigenvalue $\langle\lambda_1\rangle$ with the analytical Bjorken scaling $(\tau_0/\tau)^{2/3}$ indicated by the dotted curve. (b) Mean transverse eigenvalue $\langle\lambda_2\rangle$. (c) Spectral ratio $\langle\lambda_2/\lambda_1\rangle$ on logarithmic scale.

## 4 Neighborhood Size $k$-Sweep and Plateau Selection

When estimating local dimension on an ensemble collapsing onto a curved one-dimensional attractor, the dependence of $\langle d_{\mathrm{PR}}(k)\rangle$ on the neighborhood size $k$ reflects a balance between two distinct error sources:

$$\langle d_{\mathrm{PR}}(k)\rangle \approx d_{\mathrm{true}} - \frac{C_{\mathrm{stat}}}{k} + C_{\mathrm{curv}}\,\kappa^2 \left(\frac{k}{N}\right)^2. \tag{26}$$

1. **Undersampling / Finite-sample bias (small $k$):** For $k < 6$, random configurations of nearest neighbors in 2D space systematically underestimate the lower eigenvalue $\lambda_2$, depressing $d_{\mathrm{PR}}$ toward 1.
2. **Plateau interval ($k_{\min} \le k \le k_{\mathrm{curv}}$):** Over intermediate neighborhood sizes, the finite-sample bias decays while the neighborhood diameter remains smaller than the local radius of curvature $R = 1/\kappa$. In this region, $\langle d_{\mathrm{PR}}(k)\rangle$ exhibits a plateau.
3. **Curvature contamination (large $k$):** For large $k$, the neighborhood spans an extended arc of the curved attractor curve $\mathcal{A}_{\mathrm{attr}}(w)$. The extrinsic sagitta introduces an apparent normal variance $\lambda_2^{\mathrm{curv}} \propto \kappa^2 (k/N)^4$, which inflates $\langle d_{\mathrm{PR}}(k)\rangle$.

### 4.1 Inflection Point Plateau Detection Algorithm

To locate the center of the plateau automatically without manual parameter tuning, we compute the finite step difference:

$$\Delta(k) \equiv \Big|\langle d_{\mathrm{PR}}(k+1)\rangle - \langle d_{\mathrm{PR}}(k)\rangle\Big|, \qquad k \in [k_{\min},\, k_{\max}-1], \tag{27}$$

with $k_{\min} = 7$. The optimal plateau neighborhood $k^*$ is identified as the local minimum of $\Delta(k)$:

$$k^* = \arg\min_{k \geq k_{\min}} \Delta(k). \tag{28}$$

### 4.2 Empirical $k$-Sweep Results

Figure 3 displays the sweep of $\langle d_{\mathrm{PR}}(k)\rangle$ (top panels) and the step difference $\Delta(k)$ (bottom panels) across $k \in [3, 50]$ evaluated at three representative proper times: $\tau = 0.20\,\mathrm{fm}/c$, $\tau = 0.55\,\mathrm{fm}/c$, and $\tau = 2.00\,\mathrm{fm}/c$. The detected plateau points $k^*$ are indicated by diamonds and vertical reference lines.

- At $\tau = 0.20\,\mathrm{fm}/c$, the cloud is space-filling (2D), and $\langle d_{\mathrm{PR}}(k)\rangle$ increases monotonically toward $\approx 1.92$. The plateau minimum occurs at $k^* = 48$ with $\Delta(k^*) = 5.42 \times 10^{-4}$ and $d_{\mathrm{PR}} = 1.9165$.
- At $\tau = 0.55\,\mathrm{fm}/c$, the system is in the hydrodynamizing regime. A clear inflection plateau appears:
  - Variant 1: $k^* = 14$, $d_{\mathrm{PR}}(k^*) = 1.6255$, $\Delta(k^*) = 1.74 \times 10^{-4}$.
  - Variant 2: $k^* = 12$, $d_{\mathrm{PR}}(k^*) = 1.5986$, $\Delta(k^*) = 1.08 \times 10^{-3}$.
  - Variant 3: $k^* = 16$, $d_{\mathrm{PR}}(k^*) = 1.6427$, $\Delta(k^*) = 1.72 \times 10^{-3}$.
  - Per-slice $Z$-score: $k^* = 18$, $d_{\mathrm{PR}}(k^*) = 1.7513$, $\Delta(k^*) = 1.85 \times 10^{-4}$.
- At $\tau = 2.00\,\mathrm{fm}/c$, the cloud is collapsed onto the 1D attractor. The plateau gives $d_{\mathrm{PR}} = 1.0003$ (Var 1), 1.0002 (Var 2), 1.0000 (Var 3), and 1.0001 ($Z$-score), with step differences $\Delta(k^*) < 1.5 \times 10^{-5}$.

## 5 Dimensional Trajectories Across Proper Time

Figure 4 displays the time evolution of the ensemble-averaged Participation Ratio dimension $\langle d_{\mathrm{PR}}(\tau)\rangle$ across proper times $\tau \in [0.20, 3.00]\,\mathrm{fm}/c$.

- **Panel (a) (Fixed Neighborhood $k = 16$):** At early times ($\tau \leq 0.35\,\mathrm{fm}/c$), all methods yield $\langle d_{\mathrm{PR}}\rangle \approx 1.82$. During the collapse phase ($\tau \in [0.45, 1.00]\,\mathrm{fm}/c$), Variant 1, Variant 2, and Variant 3 decay rapidly (e.g., reaching 1.59–1.64 at $\tau = 0.55\,\mathrm{fm}/c$), whereas the per-slice $Z$-score benchmark remains higher at 1.7480. For $\tau \geq 2.00\,\mathrm{fm}/c$, all methods converge asymptotically to 1.00.
- **Panel (b) (Adaptive Neighborhood $k^*(\tau)$):** Evaluating $d_{\mathrm{PR}}$ at the adaptive plateau value $k^*(\tau)$ determined by Eq. (28) yields $\langle d_{\mathrm{PR}}\rangle \approx 1.90$ at $\tau_0 = 0.20\,\mathrm{fm}/c$, decays through the inflection values at intermediate times, and terminates at 1.0000 on the attractor.

## 6 Tabulated Numerical Summary

Table 1 compiles the exact numerical values obtained from the simulation runs across representative proper times $\tau$ for all four normalization schemes.

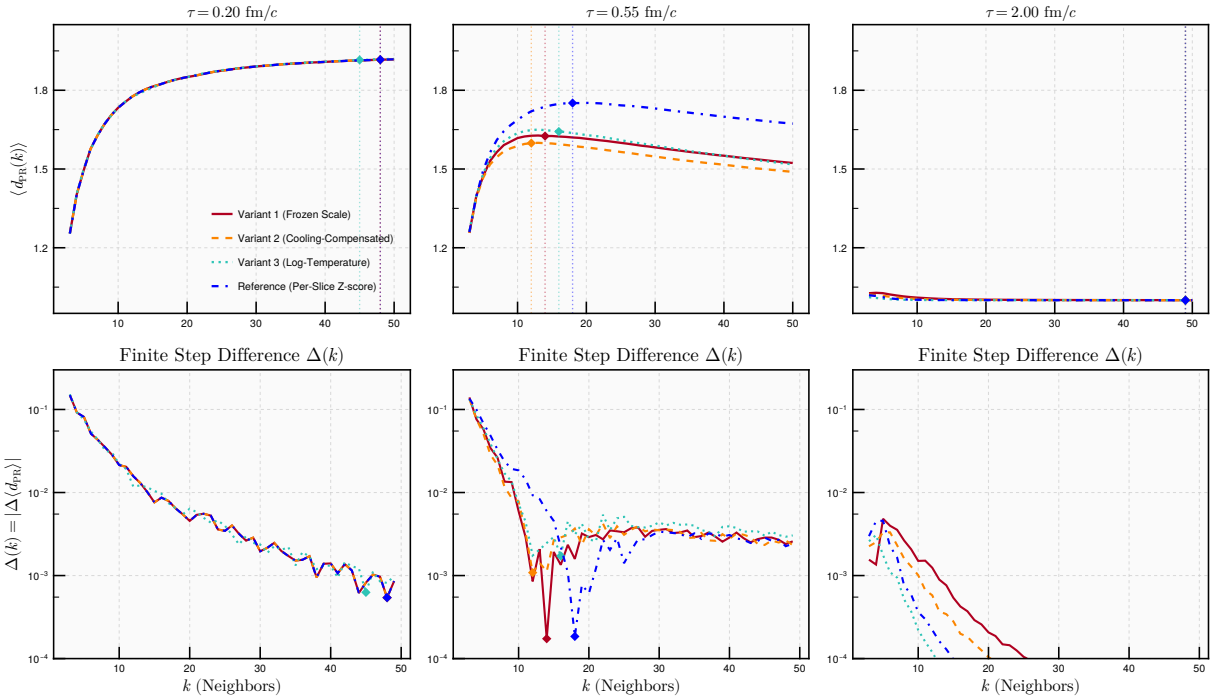



Figure 3: **Neighborhood size $k$-sweep and automated plateau detection.** Top panels: $\langle d_{\mathrm{PR}}(k) \rangle$ for $k \in [3, 50]$ at proper times $\tau = 0.20\,\mathrm{fm}/c$, $\tau = 0.55\,\mathrm{fm}/c$, and $\tau = 2.00\,\mathrm{fm}/c$. Bottom panels: Step difference $\Delta(k) = |\langle d_{\mathrm{PR}}(k+1) \rangle - \langle d_{\mathrm{PR}}(k) \rangle|$. Diamond markers denote the detected plateau minima $k^*$.

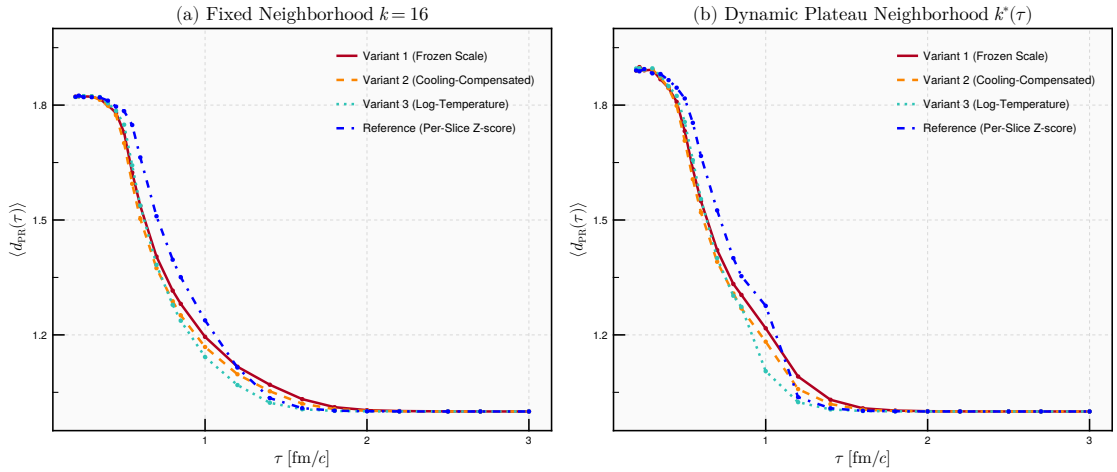



Figure 4: **Dimensional trajectories $\langle d_{\mathrm{PR}}(\tau) \rangle$ across proper time.** (a) Evaluated at fixed neighborhood size $k = 16$. (b) Evaluated dynamically at the adaptive plateau neighborhood $k^*(\tau)$.

Table 1: **Numerical summary of local PCA and participation ratio dimension across proper time $\tau$.**

| $\tau$ [fm/$c$] | Method | $\langle d_{\mathrm{PR}}(k=16)\rangle$ | $k^*$ | $\langle d_{\mathrm{PR}}(k^*)\rangle$ | $\langle\lambda_1\rangle$ | $\langle\lambda_2\rangle$ | $\langle\lambda_2/\lambda_1\rangle$ |
|---|---|---|---|---|---|---|---|
| 0.200 | Var 1 (Frozen) | 1.8217 | 34 | 1.8993 | $1.884\times10^{-3}$ | $1.034\times10^{-3}$ | 0.5701 |
| | Var 2 (Compensated) | 1.8217 | 34 | 1.8993 | $1.884\times10^{-3}$ | $1.034\times10^{-3}$ | 0.5701 |
| | Var 3 (Log $T$) | 1.8211 | 31 | 1.8935 | $2.056\times10^{-3}$ | $1.124\times10^{-3}$ | 0.5699 |
| | Per-Slice $Z$-score | 1.8217 | 34 | 1.8993 | $1.884\times10^{-3}$ | $1.034\times10^{-3}$ | 0.5701 |
| 0.350 | Var 1 (Frozen) | 1.8158 | 32 | 1.8737 | $1.530\times10^{-4}$ | $8.317\times10^{-5}$ | 0.5642 |
| | Var 2 (Compensated) | 1.8139 | 32 | 1.8714 | $1.763\times10^{-4}$ | $9.570\times10^{-5}$ | 0.5628 |
| | Var 3 (Log $T$) | 1.8210 | 30 | 1.8762 | $2.701\times10^{-4}$ | $1.469\times10^{-4}$ | 0.5701 |
| | Per-Slice $Z$-score | 1.8200 | 34 | 1.8879 | $6.344\times10^{-4}$ | $3.431\times10^{-4}$ | 0.5696 |
| 0.550 | Var 1 (Frozen) | 1.6238 | 14 | 1.6255 | $2.198\times10^{-5}$ | $1.126\times10^{-5}$ | 0.4129 |
| | Var 2 (Compensated) | 1.5945 | 12 | 1.5986 | $2.924\times10^{-5}$ | $1.476\times10^{-5}$ | 0.3933 |
| | Var 3 (Log $T$) | 1.6427 | 16 | 1.6427 | $5.341\times10^{-5}$ | $2.740\times10^{-5}$ | 0.4227 |
| | Per-Slice $Z$-score | 1.7480 | 18 | 1.7513 | $1.553\times10^{-4}$ | $7.881\times10^{-5}$ | 0.5021 |
| 0.850 | Var 1 (Frozen) | 1.2810 | 11 | 1.2924 | $3.635\times10^{-6}$ | $1.402\times10^{-6}$ | 0.1824 |
| | Var 2 (Compensated) | 1.2510 | 9 | 1.2663 | $6.141\times10^{-6}$ | $2.089\times10^{-6}$ | 0.1613 |
| | Var 3 (Log $T$) | 1.2369 | 9 | 1.2615 | $1.080\times10^{-5}$ | $4.281\times10^{-6}$ | 0.1492 |
| | Per-Slice $Z$-score | 1.3506 | 10 | 1.3613 | $3.626\times10^{-5}$ | $1.587\times10^{-5}$ | 0.2253 |
| 1.000 | Var 1 (Frozen) | 1.1949 | 9 | 1.2107 | $2.071\times10^{-6}$ | $5.871\times10^{-7}$ | 0.1239 |
| | Var 2 (Compensated) | 1.1684 | 11 | 1.1818 | $4.069\times10^{-6}$ | $9.258\times10^{-7}$ | 0.1060 |
| | Var 3 (Log $T$) | 1.1423 | 9 | 1.1703 | $6.617\times10^{-6}$ | $1.926\times10^{-6}$ | 0.0867 |
| | Per-Slice $Z$-score | 1.2379 | 10 | 1.2629 | $2.250\times10^{-5}$ | $8.081\times10^{-6}$ | 0.1478 |
| 2.000 | Var 1 (Frozen) | 1.0035 | 34 | 1.0007 | $7.444\times10^{-7}$ | $1.886\times10^{-9}$ | $1.759\times10^{-3}$ |
| | Var 2 (Compensated) | 1.0017 | 34 | 1.0004 | $2.707\times10^{-6}$ | $2.686\times10^{-9}$ | $8.702\times10^{-4}$ |
| | Var 3 (Log $T$) | 1.0004 | 34 | 1.0001 | $4.321\times10^{-6}$ | $3.086\times10^{-9}$ | $1.972\times10^{-4}$ |
| | Per-Slice $Z$-score | 1.0006 | 34 | 1.0001 | $1.886\times10^{-5}$ | $2.180\times10^{-8}$ | $3.143\times10^{-4}$ |
| 3.000 | Var 1 (Frozen) | 1.0000 | 34 | 1.0000 | $5.509\times10^{-7}$ | $7.125\times10^{-12}$ | $8.000\times10^{-6}$ |
| | Var 2 (Compensated) | 1.0000 | 34 | 1.0000 | $2.650\times10^{-6}$ | $9.901\times10^{-12}$ | $2.848\times10^{-6}$ |
| | Var 3 (Log $T$) | 1.0000 | 34 | 1.0000 | $4.210\times10^{-6}$ | $9.949\times10^{-12}$ | $5.511\times10^{-7}$ |
| | Per-Slice $Z$-score | 1.0000 | 31 | 1.0000 | $1.894\times10^{-5}$ | $1.221\times10^{-10}$ | $1.312\times10^{-6}$ |